



CESIM TEACHING NOTE

Logistics Microsimulation

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1. Simulation Overview

The Logistics Microsimulation is designed to enhance learning by focusing on the development of data-driven, client-centric strategies for optimizing multimodal transportation networks. It builds participants' capabilities in strategic logistics planning and execution, enabling them to effectively balance cost, time, environmental impact, and service quality. The simulation emphasizes informed decision-making in mode selection, route optimization, and turnaround time (TAT) management, while highlighting the importance of evaluating trade-offs and aligning logistics operations with the evolving needs of clients in a dynamic supply chain environment.

Learning Objectives

- Optimize multimodal transportation networks balancing cost, time, environmental impact, and service quality
- Make data-driven decisions on transportation mode selection, route planning, and technology investments
- Develop strategies to meet turnaround time (TAT) requirements while minimizing logistics costs
- Analyze trade-offs between different transport options (truck types, rail, air) considering their unique characteristics
- Implement appropriate inventory management policies based on client requirements

Target Audience

- Graduate and undergraduate business students in logistics, supply chain management, and operations
- Professional development programs for logistics managers and supply chain professionals
- Corporate training for transportation planners and logistics decision-makers

Time Required

- Brief introduction and setup: 20-30 minutes
- Simulation gameplay: 60-90 minutes (can be extended across multiple sessions)
- Debrief and discussion: 30-45 minutes

Case Scenario and Winning Criteria

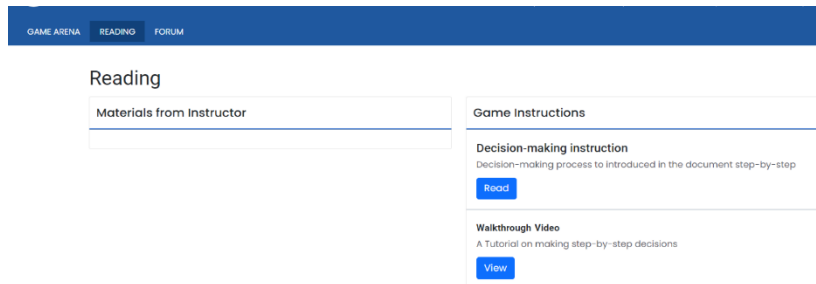
Participants in this simulation take on the role of strategic logistics managers, handling operations for four clients with diverse needs—FMCG (timely restocking), FMCD (speed-focused), textiles (flexibility), and e-commerce (temperature-controlled, time-sensitive deliveries). The goal is to create tailored, cost-effective logistics plans that ensure service quality and operational feasibility. Success is measured by how well participants meet client-specific requirements, minimize turnaround time (TAT) breaches, and maintain a cost-to-revenue ratio of **20% or lower**.

Simulation Structure

- The microsimulation is organized into independent rounds, each presenting either consistent scenarios or allowing the instructor to customize market conditions to simulate decision-making challenges throughout the company's growth journey.

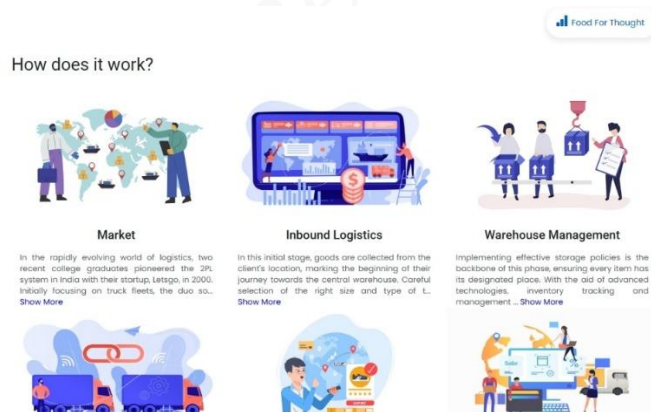
2. Pre - Simulation Preparation

a. Review Background Materials



- o Login as participant to explore the platform
- o Read the decision-making guide thoroughly
- o Watch the walkthrough video for practical insights
- o Understand the company context and business objectives

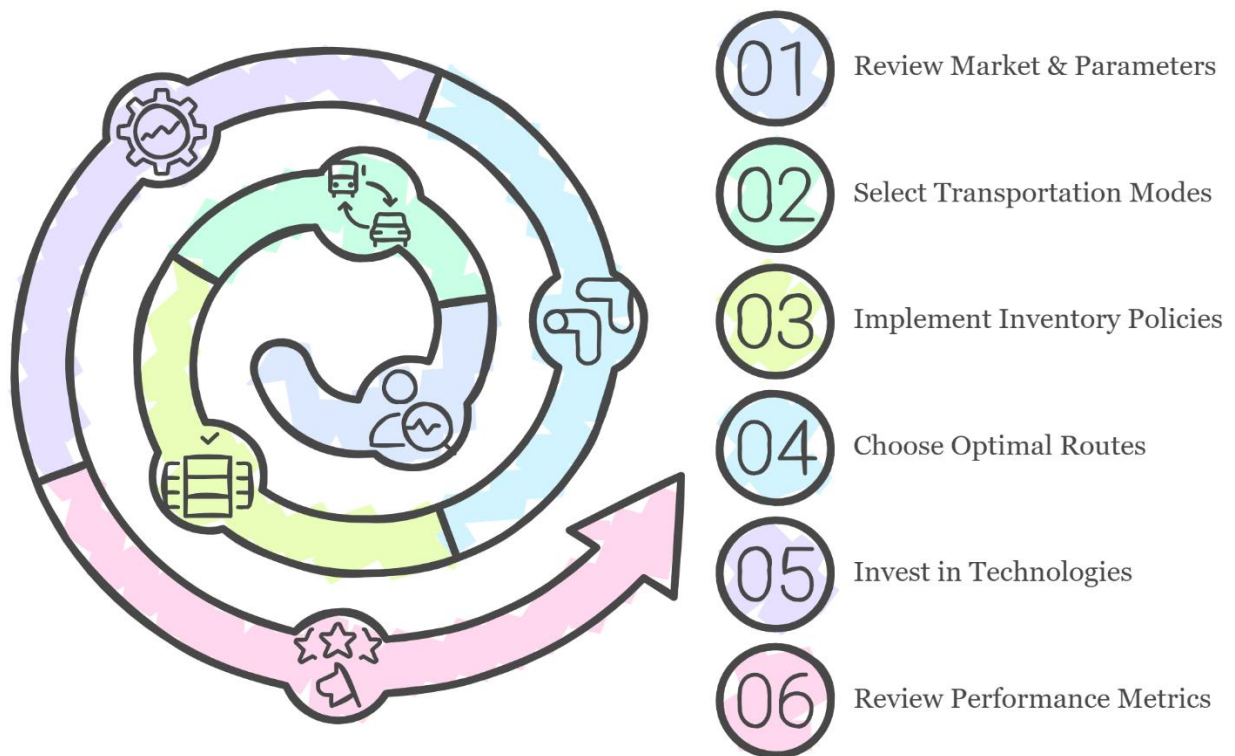
b. Familiarize with the Interface



- o Explore the Game Arena layout
- o Understand the sequence of decision tabs
- o Review the "Food for Thought" section for critical considerations

c. Understanding Decision-Making Process

The simulation follows a logical step-by-step process. Participants should move through each stage to make thoughtful decisions.



The simulation uses a multi-variable algorithm that calculates:

- Total logistics cost based on distance, fuel consumption, wages, and technology investments
- Turnaround time based on speed, distance, stoppage requirements, and route quality
- Emissions based on vehicle type, distance, and fuel efficiency factors
- Revenue based on transport mode pricing and volume
- Effectiveness percentage as the primary performance metric

e. Key Instructor-Only Parameters

Backend parameters instructors should know:

- **TAT Penalty Factor:** 0.3 (applies to late deliveries)
- **Underutilization Opportunity Cost:** ₹15,000 per vehicle
- **Emission Factors:** CO₂ (2.68 kg/l for diesel, 1.96 for air), CH₄ (0.005 g/l for diesel), N₂O (0.007 g/l for diesel)
- **Upgrade Multiplier:** 1.41 (applies to technology investments)
- **Client-Specific FIFO/LIFO Multipliers:** Range from 0.97 to 1.05
- **Technology Impact Coefficients:** Range from 0.85 to 1.12 across different parameters
- **Route Quality Factors:** Delhi-Mumbai-Bangalore (1.01), Delhi-Ahmedabad-Bangalore (0.98), Delhi-Indore-Bangalore (1.06)

f. Teaching with this Information

Instructors can leverage backend parameters to highlight why certain strategies outperformed others, often revealing counter-intuitive optimal solutions. These insights help demonstrate the

importance of balancing multiple factors and how small decisions can significantly impact overall logistics performance, reflecting the complexity of real-world decision-making.

g. Common Participant Challenges

- Overinvesting in expensive technologies without proper cost-benefit analysis
- Underestimating the impact of route selection on overall performance
- Failing to match transportation modes to client requirements
- Overlooking the importance of turnaround time commitments
- Neglecting environmental considerations when optimizing costs

Instructor Pre-Implementation Checklist for Simulation Delivery

To ensure a smooth and impactful simulation experience for your learners, please review and complete the following preparatory steps:

- **Familiarize Yourself with the Simulation:** Complete a test run of the simulation to understand its structure, decision flow, and pacing. This allows you to better anticipate participant questions and challenges.
- **Review Simulation Mechanics:** Examine the instructor-only backend settings and key variables (e.g., cost structures, demand drivers, elasticity assumptions) to understand how decisions influence outcomes.
- **Align with Course Objectives:** Identify the core concepts from your course that align with the simulation to reinforce key learning goals.
- **Prepare Contextual Examples:** Bring 2–3 real-world examples from relevant industries to illustrate successful and unsuccessful strategic decisions during the debrief.
- **Review Instructor Insights and Scoring:** Familiarize yourself with the instructor account, leaderboard and performance indicators available to instructors. These tools will help guide feedback and encourage reflective learning.
- **Ensure Participant Access:** Confirm that all participants have registered using the right link / code and are able to access the simulation platform before the session begins.
- **Test Technical Readiness:** Coordinate with your IT team (if applicable) to ensure all devices, software, and internet connectivity are ready and compatible with the simulation requirements.

3. Facilitation Roadmap

Pre-Simulation Briefing

- a. Introduce the case scenario available in the market tab.

Summary of the default Case:

Letsgo, a fast-growing logistics provider, is managing key transport operations from **Delhi to Bangalore**, a major supply chain corridor. In this scenario, the **participant acts as a Logistics Consultant**, responsible for delivering efficient, cost-effective, and client-focused logistics solutions.

The participant manages four distinct clients:

- **Unison Limited Client 1 (FMCG)** – Needs timely restocking.
- **Prompton Incorporation Client 2 (FMCD)** – Prioritizes rapid delivery.
- **Fix Corporate Client 3 (Textiles)** – Requires flexible, dependable logistics.
- **eCom Ltd. Client 4 (E-commerce)** – Demands temperature-controlled, time-sensitive deliveries.

The objective is to develop **customized logistics plans** that fulfill each client's requirements while maintaining **service-level agreements** and delivering **operational excellence**. The **simulation winning criteria** focus on both **cost-efficiency and performance quality**. Participants must aim for the **lowest logistics effectiveness percentage** ($\text{Total Logistics Cost} \div \text{Total Revenue}$), with an ideal target of **20 % or lower**.

- b. Explain the simulation flow
- c. Clarify the distinction between white decision fields (active choices) and blue decision fields (estimation cells)
- d. Set expectations: decisions should be data-driven rather than based on personal gaming preference.

During Simulation

1. First round guidance:

- Recommend a conservative approach for initial decisions
- Suggest focusing on understanding the relationship between decisions and outcomes
- Encourage note-taking on results for future rounds

2. Mid-simulation interventions:

- If participants struggle with specific concepts, provide targeted clarification
- Consider introducing "market disruptions" to test adaptability
- Prompt reflection on the implications of performance metrics

3. Final rounds:

- Encourage optimization of strategies based on accumulated learning
- Remind participants to ensure TAT compliance
- Suggest considering environmental impact alongside cost metrics

Debrief Structure

1. Performance Distribution

- Display anonymized class results showing the range of outcomes
- Identify what distinguishes top performers from others

2. Strategy Alignment Discussion

- Compare well-aligned versus misaligned strategies from the class
- Show how strategic coherence manifested in specific performance outcomes

3. Key Decision Analysis

- Review optimal approaches for critical decision areas
- Share the backend data explaining why certain choices performed better

4. Theoretical Connections

- Highlight how various concepts manifested in the simulation
- Connect simulation outcomes to theoretical principles

5. Real-World Applications

- Relate simulation insights to actual industry practices
- Discuss how these principles apply in broader contexts

4. Critical Decision Points Analysis

Decision Point 1: Transportation Mode Selection

Key Decision

Participants are required to select the most suitable mode of transportation for both inbound and outbound logistics. The optimal number and type of vehicles—across Small Trucks, Open Body Trucks, Covered Containers, Rail, and Air—should be determined based on client requirements, cargo characteristics, and the cost parameters provided.

Additionally, participants must decide whether to apply optimization techniques to these vehicles and how to allocate them effectively across different client needs.

Optimal Approach

Analyse the unique characteristics of each mode

- **Small Trucks:** One small truck has 4 MT capacity, average speed 50kmph, medium damage risk
- Minimum 10 to maximum 35 trucks can be used.
- **Open Body Trucks:** Medium capacity, one truck has 6.5 MT Capacity, average speed 45 kmph, highest damage risk.
- **Covered Container:** Highest capacity, one truck has 8.5 MT capacity, average speed 40 kmph, lowest damage risk.
- **Rail:** Very high capacity (100MT per trip), slow speed (58 kmph), high damage risk
- **Air:** Low capacity (20MT), extremely fast (750 kmph), low damage risk
- **TAT Penalty Factor:** 0.3
- Underutilization Opportunity Cost, INR:15000
- Each litre of diesel emits 2.68 kg of CO₂, plus trace amounts of potent greenhouse gases CH₄ and N₂O, underscoring the need for fuel-efficient, sustainable logistics through optimized routing, multi-modal transport, and greener technologies.

Mode-Client Match:

Align transportation mode with cargo characteristics, urgency, and cost-efficiency to ensure optimal logistics performance. Use Covered Containers or Air for fragile or high-value goods, Air for time-sensitive or temperature-controlled shipments, and Rail for cost-effective movement of bulk, low-fragility cargo.

Common Pitfalls

- Over-reliance on a single transport mode
- Ignoring damage risk factors when selecting modes
- Failing to consider lead time in mode selection
- Underutilizing high-capacity options for appropriate clients

Transportation Mode	Capacity (MT)per truck Inbound	Capacity (MT)per truck Outbound	Speed (kmph)	Fuel Efficiency (kmpl)	Price (₹/mt-km)	Damage Risk
Small Trucks	4	10	55	18	3.6	Medium
Open Body Trucks	6.5	18	50	16.5	3.3	High
Covered Container	8.5	23	45	14	3.4	Low
Rail	NA	100	58	20*	2.2	High
Air	NA	20	750	N/A**	12	Low

*Rail fuel measured differently (per km, not kmpl) **Air uses different fuel measure (₹156/liter)

Decision Point 2: Route Selection and Optimization

Key Decision

Choosing the optimal route among three options (Delhi-Mumbai-Bangalore, Delhi-Ahmedabad-Bangalore, Delhi-Indore-Bangalore) based on distance, road quality, stoppage requirements, and security considerations.

Optimal Approach

Analyse the characteristics of each route:

- **Delhi-Mumbai-Bangalore:** Medium distance (2350 km), medium road quality, high stoppage demand (415), medium theft risk
- **Delhi-Ahmedabad-Bangalore:** Longest distance (2600 km), high road quality, medium stoppage demand (220), low theft risk
- **Delhi-Indore-Bangalore:** Shortest distance (2200 km), low road quality, low stoppage demand (110), high theft risk.

Route	Distance (km)	Road Quality	Stoppages	Theft Risk	Fuel Consumption	Best Use Case
Delhi-Mumbai-Bangalore	2350	Medium (1.01)	High (415)	Medium (1.04)	93	Balanced option
Delhi-Ahmedabad-Bangalore	2600	High (0.98)	Medium (220)	Low (1.03)	92	Fragile goods
Delhi-Indore-Bangalore	2200	Low (1.06)	Low (110)	High (1.06)	94	Time-sensitive, robust goods

Match routes to specific needs:

- For fragile cargo, prioritize routes with high road quality (Delhi-Ahmedabad-Bangalore)
- For time-sensitive deliveries, balance distance with road quality factors
- For high-value goods, consider theft risk factors

Utilize backend parameters:

- Road quality multiplier affects fuel consumption and vehicle wear
- Stoppage disadvantage impacts delivery time
- Theft cost multiplier influences insurance costs

Common Pitfalls

- Selecting routes based solely on distance
- Ignoring road quality impacts on vehicle maintenance and fuel consumption
- Underestimating the effect of stoppages on overall delivery time
- Failing to consider security factors for high-value shipments

Decision Point 3: Routes Technology Investment Strategy

Key Decision

Selecting the optimal mix of technology investments to enhance operational efficiency, reduce costs, and improve service quality.

Optimal Approach

Evaluate ROI for each technology option:

- **Transportation Management System:** High cost (₹370,000), comprehensive benefits (efficiency 1.1, fuel efficiency 0.95)
- **TAT Visibility:** Medium cost (₹310,000), strong impact on TAT performance (efficiency 1.09)
- **GPS Positioning:** Medium-high cost (₹300,000), reduces theft (0.85) and improves efficiency (1.09)
- **Driver Behavioral System:** Medium cost (₹250,000), improves efficiency (1.12) and reduces fragility costs (0.91)
- **RFID Chips:** Lower cost (₹175,000), reduces theft (0.93) with moderate efficiency gains (1.05)
- **Fuel Trackometer:** Lowest cost (₹150,000), modest benefits in theft reduction (0.94) and efficiency (1.04)

Prioritize investments based on specific operational challenges:

- For TAT compliance issues: TAT Visibility system
- For theft/security concerns: GPS Positioning + RFID Chips
- For overall efficiency: Transportation Management System
- For driver-related issues: Driver Behavioral System
- Consider the upgrade multiplier (1.41) when evaluating long-term benefits.

Technology	Cost (₹)	Efficiency Impact	Theft Reduction	Fragility Reduction	Fuel Efficiency	Best Use Case
Transportation Management System	370,000	1.1	0.89	0.87	0.95	Comprehensive solution
TAT Visibility	310,000	1.09	0.9	0.93	0.96	TAT improvement
GPS Positioning	300,000	1.09	0.85	0.85	0.97	Security focus
Driver Behavioral System	250,000	1.12	1.0	0.91	0.98	Driver management
RFID Chips	175,000	1.05	0.93	0.97	0.995	Theft prevention
Fuel Trackometer	150,000	1.04	0.94	1.0	0.98	Fuel efficiency

Common Pitfalls

- Investing in too many technologies without proper ROI analysis
- Ignoring the compounding effects of complementary technologies
- Failing to match technology investments to specific operational challenges
- Overlooking the upgrade multiplier for long-term planning

Decision Point 4: Inventory Management Policy Selection

Key Decision

Inventory Management Policy

Effective inventory management plays a critical role in ensuring product quality, reducing handling costs, and meeting service level agreements. Choosing between FIFO (First-In, First-Out) and LIFO (Last-In, First-Out) policies should be based on product fragility, perishability, and handling complexity.

Policy Selection Guidelines

- LIFO is more suitable for high-fragility or sensitive products, as it reduces handling frequency by retrieving the most recently stored items. This minimizes the risk of damage and benefits from higher efficiency multipliers.
- FIFO is preferred for perishable goods, time-sensitive inputs, or items with a short shelf life, ensuring that older stock is dispatched first to maintain freshness and reduce spoilage.

By analysing backend efficiency multipliers, operations can be aligned to maximize handling efficiency:

- Fragile or delicate goods typically align with LIFO due to reduced movement and better preservation.
- Perishables and fast-moving consumer items align with FIFO, emphasizing turnover and freshness.

Dock Allocation Strategy

Given limited warehouse dock availability, allocating docks based on inbound and outbound volume, as well as service urgency, is essential for optimal flow.

Allocation Considerations

- High-volume products require proportionally more docks to prevent bottlenecks during peak loading/unloading times.
- Products with short turnaround time (TAT) requirements should be prioritized for dock access to ensure on-time dispatch.
- A balanced allocation that aligns with both volume and urgency enhances overall warehouse throughput and minimizes congestion.

Optimal Approach

Analyse segment-specific needs and backend multipliers:

- **Client A (high fragility goods):** Benefits from **LIFO** due to reduced handling; supported by a 1.05 efficiency multiplier.
- **Client B (time-sensitive industrial inputs):** Also suited to **LIFO** for faster access to newer stock; multiplier of 1.05 enhances responsiveness.
- **Client C (durable goods like textiles):** Best served by **FIFO**, ensuring orderly movement with a 1.04 efficiency gain.
- **Client D (perishable or fast-moving goods):** Strongly benefits from **FIFO** to maintain freshness and quality; multiplier of 1.03 supports turnover speed.

Optimize dock allocations based on volume and urgency:

- Higher volume segments (e.g., Clients A and D with orders above 600 MT) should receive greater dock allocation to manage throughput effectively.
- Shorter turnaround time (TAT) requirements (e.g., Client D with a 5-day TAT) should be prioritized for dock access to meet service expectations.
- Dock utilization must be balanced to prevent congestion, maintain flow, and align with the 18-dock capacity constraint.

Common Pitfalls

- Applying the same inventory policy across all clients
- Ignoring the mathematical impact of policy multipliers
- Under allocating docks to high-volume or time-sensitive clients
- Failing to match warehouse technologies to inventory policies

Client	Order Volume (MT)	Fragility	TAT Requirement (Days)	Optimal Policy	Policy Multiplier	Revised Dock Allocation
Client A	875	High	8	LIFO	1.05	5 docks
Client B	645	High	9	LIFO	1.05	5 docks
Client C	390	Low	12	FIFO	1.04	3 docks
Client D	612	Medium	5	FIFO	1.03	5 docks
Total						18 docks

Decision Point 5: Warehouse Technology Investment

Key Decision

Selecting appropriate warehouse technologies to optimize operations, reduce errors, and enhance throughput.

Optimal Approach

Evaluate each technology based on specific warehouse needs:

- **Warehouse Management System:** Comprehensive solution improving productivity (1.05), efficiency (1.08), error reduction (0.93)
- **Shuttle System:** High cost (₹400,000) but strong productivity gains (1.09) and efficiency (1.04)
- **Stacker Cranes:** Medium-high cost (₹280,000) with good efficiency (1.1) and error reduction (0.92)
- **Conveyor System:** Medium cost (₹230,000) with balanced benefits (productivity 1.05, efficiency 1.05)
- **Stretch Wrappers:** Lower cost (₹220,000) with moderate error optimization (0.91)

Prioritize investments based on warehouse challenges:

- For high-volume operations: Shuttle System + Conveyor System
- For fragile goods handling: Stretch Wrappers + Stacker Cranes
- For comprehensive improvement: Warehouse Management System
- Consider the long-term impact of the upgrade multiplier (1.41).

Common Pitfalls

- Overinvesting in warehouse technology without volume justification
- Selecting technologies that don't address specific operational bottlenecks
- Ignoring the energy consumption impacts of warehouse technologies
- Failing to align warehouse investments with transportation strategies

Technology	Cost (₹)	Productivity	Efficiency	Error Optimization	Energy Consumption Reduction	Best Use Case
Warehouse Management System	380,000	1.05	1.08	0.93	0.95	Overall optimization
Shuttle System	400,000	1.09	1.04	0.98	0.96	High volume warehouses
Stacker Cranes	280,000	1.0	1.1	0.92	0.99	Vertical storage

						optimization
Conveyor System	230,000	1.05	1.05	1.0	0.97	Throughput improvement
Stretch Wrappers	220,000	1.0	1.0	0.91	0.96	Fragile goods protection

5. Understanding Decision Linkages and System Dynamics

The logistics microsimulation represents a complex system where decisions in one area significantly impact outcomes in others. The following diagram illustrates these key relationships:

Key System Relationships

1. Mode-Route Interaction:

- Higher quality routes reduce the impact of vehicle damage risks
- Route distance directly affects fuel consumption by different vehicle types
- Stoppage requirements impact different modes disproportionately

2. Technology-Performance Linkages:

- Technology investments create multiplicative effects rather than additive ones
- Combined technologies can create synergistic benefits (e.g., GPS + Fuel Trackometer)
- Technology impact is proportional to fleet size and utilization

3. Inventory-Client Relationships:

- Inventory policies have different impacts based on client product characteristics
- Dock allocation efficiency correlates with order volume and TAT requirements
- FIFO/LIFO decisions have quantifiable impact multipliers in the backend.

4. Environmental-Operational Trade-offs:

- Lower emission options often incur higher operational costs
- Fuel efficiency improvements have compound benefits (cost and emissions)
- Technology investments can offset environmental impacts of transportation choice.

Summary of Core Decision Areas & Key Parameters

Decision Area	Key Parameters	Backend Multipliers	Performance Impact
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Transportation Mode	Capacity, speed, efficiency, damage risk	Damage chances (Low/Medium/High), Fuel efficiency (kmpl)	Cost, TAT, damage rate
Route Selection	Distance, road quality, stoppages, theft risk	Road quality (0.98-1.06), Theft cost (1.03-1.06)	TAT, security, fuel consumption
Technology Investment	Cost, efficiency impact, theft reduction	Upgrade multiplier (1.41), Efficiency factors (0.85-1.12)	Overall effectiveness, specialized improvements
Inventory Management	Policy type, dock allocation	Policy multipliers (0.97-1.05)	Client satisfaction, warehouse efficiency
Warehouse Operations	Technology selection, energy usage	Productivity (0.7-1.09), Error rates (0.91-1.5)	Throughput, accuracy, energy consumption

6. Performance Analysis Framework

Understanding Key Performance Indicators

1. Effectiveness Percentage

This primary KPI represents the ratio of revenue to total logistics costs:

- **Formula:** $(\text{Total Logistics Cost} / \text{Revenue}) * 100$
- **Target:** < 50% (excellent performance)
- **Impacted by:** All decision areas, particularly mode selection and technology investments
- **Interpretation:** Low percentages indicate more profitable operation.

2. Turnaround Time (TAT) Compliance

Measures adherence to contractual delivery timeframes:

- **Formula:** $(\text{Actual TAT} / \text{Contractual TAT}) * 100$
- **Target:** <100% (ahead of schedule)
- **Impacted by:** Route selection, transportation mode, technology investments
- **Interpretation:** Values over 100% indicate delays; TAT penalty factor (0.3) applies to costs

3. Emissions Profile

Quantifies environmental impact across three key gases:

- **CO2:** Main greenhouse gas, primarily from fuel combustion
- **CH4:** Methane emissions from fuel production and distribution
- **N2O:** Nitrous oxide from combustion processes
- **Impacted by:** Mode selection, fuel efficiency, technology investments
- **Interpretation:** Lower values indicate better environmental performance

4. Cost Structure Analysis

Breaks down total costs into component categories:

- **Fuel Costs:** Directly tied to distance, vehicle type, and fuel efficiency
- **Wage Costs:** Driver and helper expenses based on trip requirements
- **Technology Costs:** Monthly expenses for adopted technologies
- **Underutilization Costs:** Penalty for inefficient resource allocation
- **Other Costs:** Including TAT penalties, damage, and theft expenses

5. Revenue Components

Analyses income sources by transportation mode:

- **Truck Revenue:** Based on mt-km pricing (₹3.3-3.6 per mt-km)
- **Rail Revenue:** Lower rate (₹2.2 per mt-km) but higher volume
- **Air Revenue:** Premium rate (₹12 per mt-km) for time-sensitive cargo
- **Interpretation:** Modal mix strongly influences overall revenue profile

Summary of Key Performance Indicators and Interpretation

KPI	Calculation	Good Performance	Poor Performance	Key Drivers
Effectiveness %	(Logistics Cost/Revenue) *100	<30%	>50%	Mode selection, technology ROI
TAT Compliance	Actual/Contractual	<100%	>150%	Route choice, mode speed, dock allocation
CO2 Emissions	kg of CO2	<10,000 kg	>20,000 kg	Fuel consumption, mode mix
Logistics Cost	₹	<₹1,000,000	>₹2,000,000	Vehicle efficiency, distance, technology

7. Theoretical Connections by Simulation Section

Decision Section	Simulation Area	Theory	Summary
Transportation Mode	Inbound/Outbound Logistics	Modal Economics Theory	Explains the trade-offs between different transportation modes based on cost, time, and capacity considerations. The simulation applies varying cost structures for each mode that reflect real-world economics.
Route Selection	Network Optimization	Transportation Network Theory	Demonstrates how route characteristics beyond distance (road quality, security, stoppages) impact overall logistics performance. The simulation uses

			multipliers to represent these factors.
Technology Investment	Operational Efficiency	Technology Adoption Model	Illustrates how technology investments create compound returns through efficiency multipliers. The upgrade multiplier (1.41) represents the increasing returns on technology investments over time.
Inventory Management	Warehouse Operations	Inventory Theory (FIFO/LIFO)	Shows how different inventory management approaches affect performance depending on product characteristics. Policy multipliers quantify these impacts for different client types.
Client Management	Service Level	Service Quality Theory	Demonstrates the relationship between service levels (measured by TAT) and operational costs. The TAT penalty factor (0.3) represents the cost-of-service failures.
Environmental Impact	Emissions	Sustainability Trade-off Theory	Highlights the tension between operational efficiency and environmental impact. Emission factors for different modes and fuels quantify these trade-offs.

Critical Teaching Points

1. Systems Thinking in Logistics:

- The simulation demonstrates how decisions in one area cascade through the entire logistics system
- Optimization in isolation often leads to suboptimal overall performance
- Trade-offs between cost, time, and quality are quantifiable and manageable

2. Data-Driven Decision Making:

- Performance metrics provide actionable insights for continuous improvement
- Backend parameters reveal the "physics" of the logistics system
- Understanding multiplier effects is essential for strategic decision-making

3. Balancing Competing Objectives:

- Pure cost minimization often reduces service quality and increases environmental impact
- Time optimization typically increases costs but may be justified for certain clients

- Environmental considerations can be integrated into decision-making without sacrificing performance

4. **Technology as a Strategic Lever:**

- Technology investments should be evaluated based on their multiplier effects
- Different operational challenges require specific technological solutions
- The compounding effect of multiple technologies can create significant competitive advantages

5. **Client-Centric Logistics Design:**

- Different clients have quantifiably different requirements
- Service differentiation can be achieved through modal mix, route selection, and inventory policies
- Client-specific approaches yield better results than one-size-fits-all strategies

8. **Sample Optimal Solution**

It is essential to acknowledge that numerous combinations can lead participants to achieve the desired outcomes, and the provided solution serves as a reference for faculty and is not intended to be shared directly with participants.

Table Summarizing the key decision areas of the Microsimulation

Decision Areas	Input
Inbound Logistics	
Truck 1	35
Truck 2	35
Truck 3	35
Truck 1 optimization	Yes
Truck 2 Optimization	Yes
Truck 3 Optimization	Yes
Warehouse Management	
Unison Limited (Client A)	LIFO
Promoton Incorporation (Client B)	LIFO
Fix Corporate (Client C)	FIFO
eCom Limited (Client D)	FIFO
Unison Limited, docks	5
Promoton Incorporation, docks	5
Fix Corporate, docks	4

eCom Limited, docks	4
Warehouse Upgrade 1	Stacker Cranes
Warehouse Upgrade 2	Shuttle System
Routes & Technology	
Route	Delhi -Ahmedabad – Bangalore
Technology 1	Fuel Trackometer
Technology 2	TAT Visibility
Stage 2 -Outbound Logistics	
Small Trucks	35
Open body trucks	35
Covered container	50
Rail Trips	3
Air Trips	3
Small Trucks, Optimization	Yes
Open Body Trucks, Optimization	Yes
Covered Container, Optimization	Yes
Rail, Optimization	Yes
Air, Optimization	Yes

Decision Making Rationale

1. Inbound Logistics Decisions

- Unison Limited (Client A)
- Promoton Incorporation (Client B)
- Fix Corporate (Client C)
- eCom Limited (Client D)

Truck Fleet Selection

Decision: 35 units each (Truck 1, 2, 3) + Optimization

Factor	Rationale
Capacity	Covers 2,500+ MT demand, avoids ₹15,000/vehicle underutilization penalty
Flexibility	3 truck types balance cargo needs & risk diversification
Optimization	Multipliers (1.04–1.12) generate compound efficiency gains

Trade-offs:

- Higher upfront fleet investment ↔ Avoidance of penalties
- More fuel consumption ↔ Higher revenue capacity
- Tech costs ↔ Efficiency multipliers

Learning: Demonstrates capacity planning & deeper cost structure analysis.

2. Warehouse Management Decisions

Inventory Policy

- LIFO → Clients A & B
- FIFO → Clients C & D

Client	Rationale	Multiplier
A & B	Fragile goods → minimize handling	1.05
C & D	Textiles/e-commerce → freshness & turnover	1.04 / 1.03

Trade-offs:

- Client-specific optimization ↔ Operational complexity
- Handling efficiency ↔ Product quality maintenance

Dock Allocation

Decision: A = 5, B = 5, C = 4, D = 4 (Total 18)

Client	Volume (MT)	TAT	Docks
A	875	Medium	5
B	645	Medium	5
C	580	Low	4
D	400	High (5 days)	4

Rationale:

- A & B: Higher volumes → more docks
- D: Lower volume but strict TAT → priority allocation
- Full 18 docks → max throughput

Warehouse Technology

Decision: Stacker Cranes + Shuttle System

Technology	Cost	Benefits
Stacker Cranes	₹280,000	1.1 efficiency, 0.92 error reduction
Shuttle System	₹400,000	1.09 productivity, high throughput

Why Together? Multiplicative productivity + efficiency + error reduction.

Trade-offs:

- High investment (₹680,000) ↔ Long-term operational gains
- Single tech vs. synergistic combo

3. Route & Transportation Technology

Route Selection

Decision: Delhi → Ahmedabad → Bangalore (longest)

Factor	Rationale
Road quality	0.98 multiplier reduces fuel use
Security	Lowest theft risk (1.03)
Maintenance	Better road quality → lower wear & tear

Trade-off: 250km longer ↔ Lower theft, fuel efficiency, vehicle longevity.

Technology Selection

Decision: Fuel Trackometer + TAT Visibility

Tech	Cost	Benefits
Fuel Trackometer	₹150,000	Fuel efficiency (0.98), theft reduction (0.94)
TAT Visibility	₹310,000	On-time delivery focus, 1.09 efficiency

Why this Pairing? Tackles both **cost control** + **service delivery**.

Trade-offs:

- Immediate investment ↔ Compound benefits (1.41 multiplier)
- Focused impact ↔ Broad operational improvement

4. Outbound Logistics Decisions

Modal Mix

Mode	Units/Trips	Purpose	Rationale
Small Trucks	35	FMCG (Client A)	Frequent deliveries, 10MT capacity
Open Body Trucks	35	Textiles (Client C)	Flexible, cost-effective (₹3.3/mt-km)
Covered Containers	50	Fragile goods (A & B)	23MT, lowest damage risk
Rail	3 trips	Bulk textiles (C)	Cheapest option (₹2.2/mt-km), 100MT
Air	3 trips	E-commerce (D)	Fastest, meets 5-day TAT

All optimized → Efficiency multipliers applied across fleet.

Trade-offs:

- High optimization cost ↔ Consistent peak performance
- Modal balance ↔ Client service differentiation

5. Strategic Integration

Client-Centric Approach:

- A: Frequent small deliveries (FMCG)
- B: Speed focus via containers + air backup
- C: Flexible mix of rail + trucks for textiles
- D: Speed + turnover for e-commerce

Cost-Service Balance:

- <20% effectiveness penalty by maximizing revenue & minimizing penalties
- Compound benefits via multipliers

Sustainability Integration:

- Fuel-efficient tech
- Rail as greener option
- Route optimization reducing waste

Integrated Logistics Strategy Impact

End-to-End Efficiency: Optimization across both inbound and outbound flows increases delivery speed, load consolidation, and resource utilization.

Cost Control: Intelligent mode assignment (e.g., limiting air freight) ensures profitability while preserving service quality.

Client Satisfaction: Fragility, turnaround time, and dock preferences are embedded into operational planning, increasing SLA compliance and customer trust.

Technology Leverage: Real-time data from behavioral tracking and fuel monitoring enhances visibility and continuous improvement.

9. Common Pitfalls and Teaching Interventions

Pitfall 1: Over-reliance on a Single Transportation Mode

Symptoms: Defaulting to trucks for all shipments, leading to high fuel costs, missed TATs, and inefficiencies.

Intervention: Encourage participants to analyze shipment priorities and experiment with multimodal combinations (e.g., truck + rail or air) to balance cost, speed, and efficiency.

Pitfall 2: Ignoring Route Quality Factors

Symptoms: Choosing routes based only on distance, resulting in delays, product damage, or theft.

Intervention: Guide participants to incorporate route reliability, road conditions, and security ratings into their decision-making matrix—not just kilometers traveled.

Pitfall 3: Undiscriminating Technology Investments

Symptoms: Investing in all available technologies without assessing their impact or ROI, causing cash flow issues.

Intervention: Urge participants to align tech investments with specific pain points (e.g., temperature control or real-time tracking) and calculate returns before committing resources.

Pitfall 4: Misaligned Inventory Policies

Symptoms: Applying the same inventory approach (like FIFO) across all clients, leading to TAT failures and poor utilization of dock capacity.

Intervention: Prompt participants to customize inventory policies based on client needs—such as FEFO for perishables and JIT for time-sensitive categories.

Pitfall 5: Neglecting TAT Requirements

Symptoms: Frequent TAT breaches due to poor planning or excessive cost optimization.

Intervention: Encourage reverse-planning from client delivery deadlines and incorporate TAT as a key constraint during transport and inventory decisions.

Pitfall 6: Overlooking Environmental Impact

Symptoms: High emissions due to reliance on polluting transport modes and low use of green technologies.

Intervention: Highlight sustainability scoring and prompt participants to factor in emissions when choosing routes and modes. Recommend investing selectively in eco-efficient technologies.

10. Instructor Tips and Insights

Pre-Simulation Survey: Ask participants to predict client logistics needs or optimal transport choices upfront to surface assumptions that can be revisited during debrief.

Staged Decision Making: Break the simulation into phases (e.g., mode selection, route planning, tech investment) with interim feedback to emphasize the value of iterative strategy refinement.

Real-World Comparisons: Share 2–3 real logistics case studies (e.g., Amazon, Delhivery, Maersk) that align with participant choices to anchor learning in practical industry examples.

Focus on One Framework: In early sessions, center discussions around a key logistics concept (e.g., TAT management, cost-to-service trade-offs) to build depth before adding complexity.

Decision Log: Have participants document their reasoning for major logistics decisions (e.g., mode mix, inventory policy) to foster reflection and clarify strategic thinking.

Peer Analysis: Post-simulation, pair participants to analyze and critique each other's logistics strategies, promoting shared learning and critical evaluation.

Highlight Transferability: Connect simulation insights to broader logistics or supply chain contexts (e.g., last-mile delivery, global sourcing) to reinforce relevance beyond the case.

Suggested Discussion Starters

1. How do you decide if a logistics technology is worth implementing, and what other factors influence this choice?
2. If you could fully serve only three clients, which would you prioritize and why?
3. How do the simulation's emission factors reflect real sustainability challenges, and how can emissions be reduced without lowering performance?
4. Why do FIFO and LIFO affect clients differently, and how do policy multipliers represent real operational impacts?
5. How do you manage the trade-off between cost optimization, TAT fulfillment, and customer satisfaction?
6. What strategies help balance cost, TAT, and service quality for specific routes?
7. What factors guide your choice of transportation and warehouse tech for inbound logistics, and how do they impact TAT and cost?
8. How do tools like RFID or automated picking improve cost-effectiveness and TAT in logistics?
9. How do you incorporate sustainability into logistics planning, and what benefits can it bring to the business?
10. Can you give examples of successful logistics collaborations or partnerships that improved cost, efficiency, or innovation?

Frequently Asked Questions (FAQs)

General Simulation Questions

Q: How many rounds should we expect to reach optimal performance?

A: Most participants show significant improvement by round 2-3, with optimal strategies typically emerging by round final rounds. The learning curve is intentionally designed to reward experimentation and adaptation.

Q: How realistic are the parameters in this simulation?

A: The parameters are based on industry benchmarks but simplified for teaching purposes. The relative relationships between parameters (e.g., modal costs, emission factors) accurately reflect real-world logistics considerations.

Strategy Questions

Q: Is there a single "best" strategy for this simulation?

A: No. While certain approaches consistently outperform others, the optimal strategy depends on how you balance competing objectives like cost, time, and environmental impact. Different prioritization can lead to multiple viable approaches.

Q: Should I invest in all technologies or be selective?

A: Selective investment typically yields better results. The upgrade multiplier (1.41) means that focused investments in complementary technologies generate better returns than spreading resources across all options.

Q: How should I balance client-specific approaches versus standardization?

A: The simulation rewards client-specific approaches through policy multipliers and TAT requirements. However, some standardization in technology and route selection can create economies of scale. The optimal balance typically involves customization in modal mix and inventory policies with standardization in technology infrastructure.

Technical Questions

Q: Why do my costs increase even when I optimize fleet allocation?

A: Check your TAT compliance. The TAT penalty factor can significantly increase costs for late deliveries. Also review your technology investments, as their costs may offset some efficiency gains in the short term.

Q: How is the effectiveness percentage calculated?

A: Effectiveness % = (Logistics Cost / Total Revenue) * 100. Revenue is calculated based on modal pricing (₹/mt-km) and volumes, while costs include fuel, wages, technology, underutilization, and penalties.

Q: How do the route quality factors influence performance?

A: Route quality factors (0.98-1.06) affect multiple aspects of performance: fuel consumption, vehicle wear, delivery time, and damage probability. A lower factor (e.g., 0.98 for Delhi-Ahmedabad-Bangalore) indicates better quality with positive impacts across these dimensions.

Conceptual Questions

Q: How does this simulation relate to real-world logistics decision-making?

A: The simulation captures the key trade-offs in logistics management: modal selection, route planning, technology investment, and service differentiation. The quantified relationships between decisions and outcomes reflect the cause-effect dynamics of actual logistics operations.

Q: What is the significance of the policy multipliers in inventory management?

A: These multipliers (0.97-1.05) represent the operational efficiency gains from matching inventory policies to product characteristics. For example, fragile goods benefit from LIFO (1.05 multiplier) as they experience less handling, while perishable items benefit from FIFO (1.03 multiplier) to ensure freshness.

Q: How should emissions be factored into decision-making? A: While emissions don't directly impact the effectiveness percentage, they represent an important sustainability consideration. Forward-thinking logistics operations balance environmental impact with cost and service metrics, anticipating future regulatory and market pressures.

Conclusion

The Logistics Optimization Microsimulation provides a comprehensive platform for exploring the complex interrelationships in transportation and logistics management. By revealing the hidden mechanics behind logistics operations through backend parameters, participants gain insights into the quantifiable impacts of their decisions.

Key takeaways include:

- Strategic alignment of conflicting objectives
- Tailoring logistics strategies to specific client needs
- Making targeted technology investments with high ROI
- Utilizing data-driven insights for decision-making based on performance metrics
- Embracing systems thinking to understand interconnected decision points

These competencies directly translate to real-world logistics challenges, where the capacity to optimize across multiple facets—cost, time, quality, and sustainability—is increasingly pivotal for achieving competitive advantage.

Supplementary Information: Parameters Summary

(Available under participant Game Arena Market section)

1. Client Order & Contractual Details

Client	Order (MT)	TAT (Days)	Fragility
Unison Ltd.	875	8	High
Prompton Inc.	645	9	High
Fix Corp.	390	12	Low
eCom Ltd.	612	5	Medium

2. Freight Rate by Mode (INR/MT-km)

Mode	Rate
Small Trucks	₹3.6
Open Body Trucks	₹3.3
Covered Containers	₹3.4
Rail	₹2.2
Air	₹12.0

3. Workforce & Operations

Parameter	Value
Max truck distance/day	400 km
Inbound round trip distance	150 km
Diesel cost per litre	₹91
Drivers available	50
Helpers available	150
Helpers per 20 MT load	5
Helpers per trip	2
Driver wage/day	₹600
Helper wage/day	₹300
Contractual wage increment	10%